



Project SAPPHIRE

Semantic and Acoustic Perceptual Holistic Integration
REtrieval

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Outline

The Problem

Our Vision and Methodology

Project Execution: Sprints 1–3

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Roadmap & Conclusion

The Problem

The Perceptual Gap

Why machines hear, but don't listen.

The Disconnect Between Machine and Human

The Core Challenge

The disconnect between objective computational analysis and the subjective, nuanced nature of human music perception.

Machines Analyze

- Physical Properties
- Frequency (Hz)
- Amplitude (dB)
- Waveforms

Humans Perceive

- Subjective Attributes
- Pitch & Loudness
- Emotion & Timbre
- Groove & Feel

Current systems are experts at analysis, but novices at perception.

Why Traditional MIR Systems Fail

1. Reliance on Flawed Labels

- Predefined genres and moods are rigid and subjective.
- Human annotators often disagree, creating an inconsistent “ground truth.”
- This imposes a **performance ceiling** on AI models.

2. Uni-Modal (Acoustic-Only) Focus

- Human listening is a **holistic, multi-modal** experience.
- Existing systems ignore context from **lyrics**, **structure**, and **vocal performance**.

Our Vision and Methodology

We model the **process of human perception**, not static labels.

- **Psychoacoustically-Informed Features**
- **Holistic Multi-Modal Integration**
- **Perceptually-Aligned Learning**

Core Methodology

1. Cross-Modal Contrastive Learning

- Learns by comparing pairs of data
- Pulls audio + text into a shared embedding space
- Avoids flawed labels

2. LLMs for Data Synthesis

- Multi-modal datasets are scarce
- LLMs generate rich, context-aware descriptions

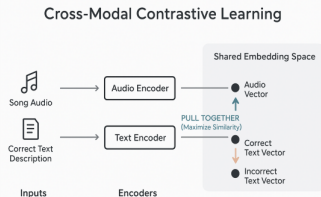


Figure 1: CMCL Architecture

Project Execution: Sprints 1–3

Data Pipeline Overview (Sprint 2)

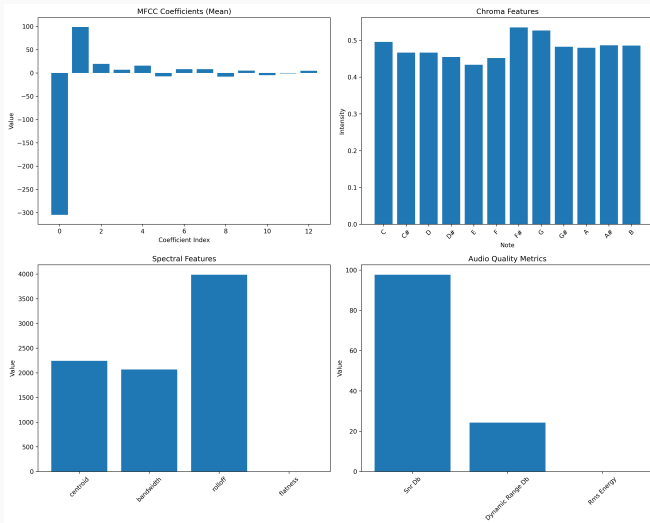
From Raw Data to Structured Corpus

- **Dataset:** Zalo2022 Lyric Alignment
- Audio resampled to 44.1 kHz, normalized to -23 LUFS
- Lyrics standardized to UTF-8
- Final corpus: **264 high-quality tracks**

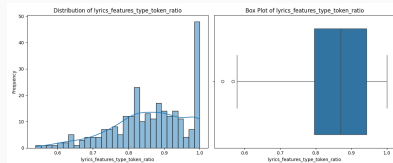
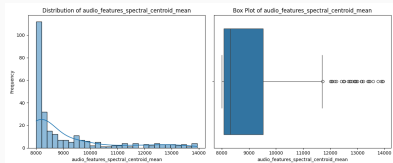
Phase 2: Multi-Modal Feature Extraction

Domain	Features
Acoustic & Timbral	MFCCs, Centroid, Bandwidth, Rolloff, RMS
Harmonic & Tonal	Chroma, Musical Key, Tonal Complexity
Rhythmic	Tempo, Regularity, Syncopation Index
Lyrical	TTR, Readability, Sentiment
Audio Quality	SNR, Dynamic Range

Feature Metric Analysis



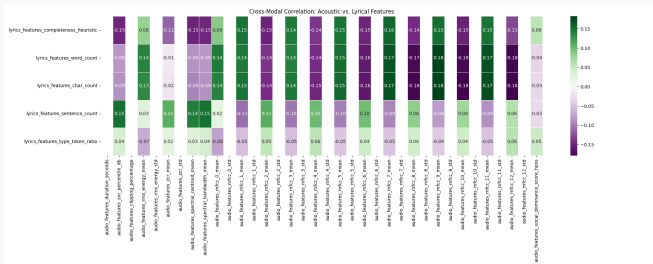
Phase 3: Exploratory Data Analysis



Key Insight

Feature distributions reveal corpus bias (brightness, lyrical complexity, etc.).

Correlation Analysis



Key Insight

Sound and meaning are only weakly correlated in raw form.

Uncovering the Latent Sonic Space (Sprint 3)

We used Principal Component Analysis (PCA) to distill the complex, high-dimensional acoustic features into their most essential axes of variation.

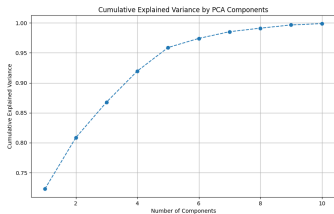


Figure 2: Cumulative Explained Variance by PCA

Core Insight

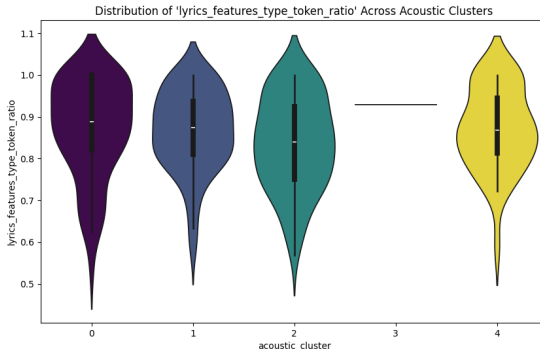
The complex “sound” of a song can be meaningfully compressed into a much lower-dimensional space, a foundational concept for our model.

Striking Result

The first two Principal Components alone capture **80.83%** of the total acoustic variance.

- **PC1:** Timbral Complexity & Density
(Bright/Rich vs. Dark/Simple)
- **PC2:** Rhythmic Energy & Percussiveness
(High-energy/Noisy vs. Smooth/Tonal)

Do Sonic Groups Share Meaning?



Key Insight

Clusters formed by sound show emerging semantic patterns.

The Result: Perceptual Gap

Average Jaccard Similarity:

0.0320

Significance

Acoustic similarity **does not** imply semantic similarity. This baseline validates the core motivation of our work.

Current Work & Next Steps

Recent Work

- Dataset expansion (Mood, Jam-alt)
- Cross-dataset feature comparison to finalize features

Dataset Expansion Details

1. JAM-ALT (Jamendo Alternative)

- **Source:** Hugging Face Datasets
- **Content:** A multi-modal dataset of music audio, lyrics, and metadata.
- **Utility:** Ideal for our contrastive learning goals, linking lyrical content directly to audio features.

2. CSD (Cross-Modal-Similarity Dataset)

- **Source:** Zenodo
- **Content:** Contains audio features and semantic annotations (tags, descriptions) for music tracks.
- **Utility:** Provides a rich source of semantic-acoustic pairs, crucial for bridging the perceptual gap.

3. MIREX MOOD (Mood Classification)

- **Source:** MIR @ DEI-UC
- **Content:** Expert-annotated dataset for mood, clustered into distinct emotional categories.
- **Utility:** Serves as a key benchmark for our mood classification task.

Putting Features to Work: Mood Classification

From Analysis to Application

Having established our feature set, we tested its practical utility on a real-world MIR task: mood classification.

The Task

Train a model to predict a song's mood based on our extracted multi-modal features.

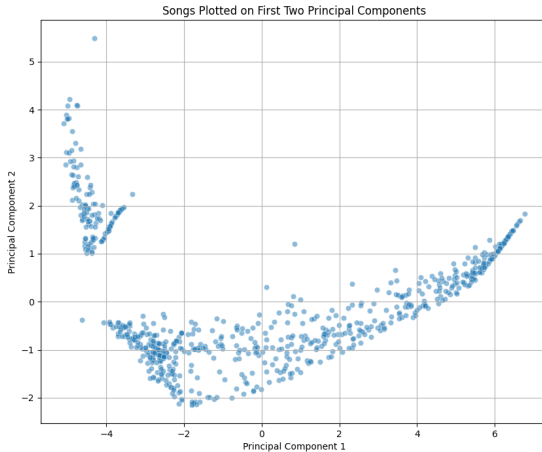
The Dataset

MIREX-like Mood Dataset, with expert annotations for 5 distinct mood clusters (e.g., Passionate/ Rousing, Bittersweet/ Wistful).

The Model

We trained a Random Forest classifier, chosen for its robustness and interpretability.

Feature Metric Analysis



Key Insights from Mood Classification

The feature importance analysis reveals what truly drives mood perception in the data.

Finding 1: Multi-Modality is Crucial

The most predictive features are a **mix of acoustic, lyrical, and rhythmic** domains. An acoustic-only model would miss critical information.

Finding 2: Energy and Emotion Lead the Way

- **RMS Energy** (perceived loudness/power) and **Spectral Centroid** (brightness) are top acoustic predictors.
- Lyrical **Sentiment** (VADER compound score) is the most important semantic feature.

Finding 3: Rhythm and Harmony Provide Nuance

Features like **Tempo** and **Tonal Complexity** are significant, confirming they contribute to the overall emotional character of the music.

Psychoacoustically-Informed Feature Selection

Timbre & Texture

- **MFCCs**
Captures the timbral “fingerprint” or color of the sound.
- **Spectral Centroid**
Correlates with the perceived “brightness” of an instrument or voice.
- **RMS Energy**
Represents the perceived loudness and intensity of the track.

Rhythm & Structure

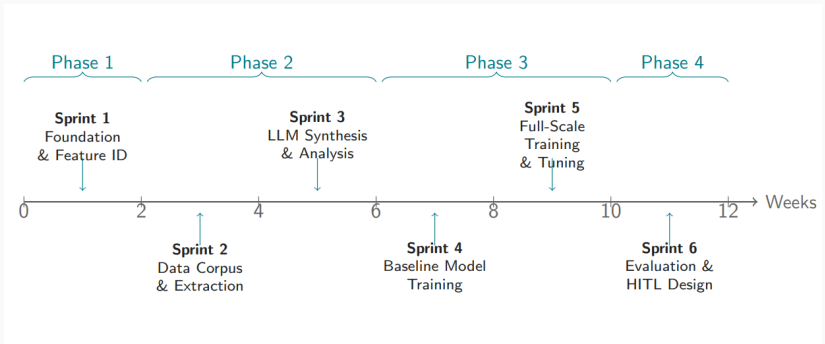
- **Tempo (BPM)**
The fundamental pace or “heartbeat” of the music.
- **Rhythm Regularity**
Measures the steadiness of the beat, influencing the “groove”.
- **Chroma Features**
Represents the tonal content (C, C#, D...), key to harmony and mood.

Semantic Content

- **Sentiment Score**
Quantifies the emotional tone of the lyrics
- **Readability Score**
Indicates lyrical complexity, from simple to abstract.
- **Semantic Embeddings**
A vector representing the core themes and meaning of the lyrics.

Roadmap & Conclusion

Project Timeline: 12-Week Plan



Conclusion

Achievements

- Robust multi-modal data pipeline
- In-depth EDA and latent analysis
- Empirical baseline: **0.0320**

Path Forward

- Finalize model using these new final features
- Contrastive learning model
- Final Product

Thank You

Questions?